

Eventual simple negative roots for rank-four capacity-two partition truncations

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7 October 2026

Version 0.1-beta preprint; not peer reviewed.

Abstract

Let $m_1 + \cdots + m_q = n$ be an arbitrary partition into positive integers and let

$$M(\mathbf{m}) = \text{trunc}_4 \left(\bigoplus_{i=1}^q U_{\min(2, m_i), m_i} \right).$$

For every fixed $0 < \alpha < 1/2$, we prove that there is an integer N_α , independent of the number of blocks and of the particular partition, such that, whenever $n \geq N_\alpha$ and $\max_i m_i \leq \alpha n$, the Ehrhart h^* -polynomial of the base polytope of $M(\mathbf{m})$ has degree $n - \lceil n/4 \rceil$ and all of its roots are distinct, real, and strictly negative. The number of blocks may grow with n , and blocks of sizes one and two are allowed.

The proof has three parts. An exact inclusion–exclusion formula reduces the Ehrhart numerator to finitely many color levels. A binomial-tail kernel is shown to have positive real part after division by $\cosh \sqrt{t}$; its even part therefore has infinitely many simple negative roots. Finally, a compactification of decreasing block masses, allowing dust, gives a threshold uniform over all partitions. The two moving endpoint regimes are controlled by a cancellation which uses $\sum_i m_i = n$, rather than the number of blocks. The threshold is not made explicit. The result does not cover the critical boundary $\alpha = 1/2$, all small orders, other ranks or capacities, or arbitrary nonpaving matroids.

1 Introduction

For a lattice polytope P of dimension d , write

$$\sum_{\ell \geq 0} |\ell P \cap \mathbb{Z}^N| z^\ell = \frac{h_P^*(z)}{(1-z)^{d+1}}.$$

We study the base polytope of a rank-four truncation of a direct sum of rank-at-most-two uniform matroids. If the ground set is partitioned into blocks $E_1, \dots, E_q$, with $|E_i| = m_i$, then its real base polytope is

$$P(\mathbf{m}) = \left\{ x \in [0, 1]^n : \sum_{e=1}^n x_e = 4, \quad \sum_{e \in E_i} x_e \leq 2 \text{ for every } i \right\}. \quad (1.1)$$

Whenever the rank of the untruncated direct sum is at least four, this is a rank-four capacity-two weighted multi-hypersimplex. In particular this condition holds throughout the eventual range of Theorem 1.1. The family includes the ordinary hypersimplex $\Delta(4, n)$ when every block has size

at most two, so it would be inaccurate to call every member of the family nonpaving. The object and several of its counting descriptions belong to established weighted multi-hypersimplex and split-matroid frameworks. Our concern is the zero set of its Ehrhart numerator, uniformly over nonuniform partitions.

Theorem 1.1. *Fix $0 < \alpha < 1/2$. There exists an integer N_α such that the following holds. For every $n \geq N_\alpha$ and every finite partition*

$$m_1 + \cdots + m_q = n, \quad m_i \geq 1, \quad \max_i m_i \leq \alpha n,$$

the polynomial $H_{\mathbf{m}}(z) = h_{P(\mathbf{m})}^(z)$ has degree*

$$D = n - \left\lceil \frac{n}{4} \right\rceil$$

and exactly D distinct roots, all lying on the negative real axis.

The threshold may depend on α , but not on q or on the partition. We do not claim an explicit value of N_α , a threshold uniform as $\alpha \uparrow 1/2$, or a statement at the critical boundary.

Adiprasito–Zhang [7, Theorem 3.1] already prove eventual simple negative real-rootedness for ordinary fixed-rank hypersimplices. Their rank-four specialization covers our uniform subfamily, for which no novelty is claimed. The increment here is a common threshold over all capacity-two partitions of bounded maximum block ratio, including nonuniform partitions and a number of blocks growing with n . This extends the object scope of that rank-four specialization, not their theorem for all fixed ranks. Section 9 records the comparison with the full text of HAL version v1.

2 Degree and exact lattice-point formulas

The first point is that small blocks do not change the eventual degree.

Lemma 2.1 (dimension and codegree). *For $n \geq 13$, under the hypotheses of Theorem 1.1, the matroid $M(\mathbf{m})$ is connected and $P(\mathbf{m})$ has dimension $n - 1$. Its codegree is*

$$c_0 = \max \left(\left\lceil \frac{n}{4} \right\rceil, \left\lceil \frac{\max_i m_i + 1}{2} \right\rceil \right). \quad (2.1)$$

Consequently, for fixed $\alpha < 1/2$ and sufficiently large n , $c_0 = \lceil n/4 \rceil$ and $\deg H_{\mathbf{m}} = n - c_0$.

Proof. Let f be the total number of elements in blocks of sizes one or two, and let c be the number of blocks of size at least three. The rank of the untruncated direct sum is $f + 2c$. The bound $\max_i m_i < n/2$ implies $f + 2c \geq 5$ when $n \geq 13$. Every pair of elements extends to a five-element independent set in the direct sum; that set becomes a circuit after rank-four truncation. Thus all elements lie in one matroid component.

An interior lattice point of $\ell P(\mathbf{m})$ satisfies

$$1 \leq x_e \leq \ell - 1, \quad \sum_e x_e = 4\ell, \quad \sum_{e \in E_i} x_e \leq 2\ell - 1. \quad (2.2)$$

The possible sum over block E_i is the full integer interval

$$[m_i, \min\{m_i(\ell - 1), 2\ell - 1\}]. \quad (2.3)$$

The lower endpoints sum to n , so $n \leq 4\ell$ is necessary. Nonemptiness of every interval is equivalent to $m_i \leq 2\ell - 1$. Conversely, once these conditions hold and $\ell \geq \lceil n/4 \rceil \geq 4$, the sum of the upper endpoints is at least 4ℓ : this follows separately from $c \geq 3$, $c = 2$, $c = 1$, and $c = 0$, using respectively $6\ell - 3$, $5\ell - 3$, $8\ell - 7$, and $n(\ell - 1)$ as lower bounds. The interval-sum property then gives (2.1). The standard relation $\deg h^* = d + 1 - \text{codeg } P$, with $d = n - 1$, completes the proof. $\square$

We next record the exact coordinate inclusion–exclusion. Put

$$C_S(\ell) = \# \left\{ y \in \mathbb{Z}_{\geq 0}^n : \sum_e y_e = (4 - |S|)\ell - |S|, \sum_{e \in E_i} y_e \leq (2 - \mathbf{1}_{i \in S})\ell - \mathbf{1}_{i \in S} \right\}. \quad (2.4)$$

Negative totals or upper bounds are interpreted as giving zero.

Lemma 2.2 (finite color inclusion–exclusion). *For every $\ell \geq 0$,*

$$L_{P(\mathbf{m})}(\ell) = \sum_{S \subseteq [q], |S| \leq 3} (-1)^{|S|} \left(\prod_{i \in S} m_i \right) C_S(\ell). \quad (2.5)$$

If

$$R_S(z) = (1 - z)^n \sum_{\ell \geq 0} C_S(\ell) z^\ell,$$

then

$$H_{\mathbf{m}}(z) = A(z) - \sum_i m_i R_i(z) + \sum_{i < j} m_i m_j R_{ij}(z) - e_3(\mathbf{m}) z^3, \quad (2.6)$$

where $A = R_\emptyset$.

Proof. Apply inclusion–exclusion to the coordinate inequalities $x_e \leq \ell$. Two selected violating coordinates cannot lie in the same block, since their block sum would exceed 2ℓ . After subtracting $\ell + 1$ from one selected coordinate in every selected block, one obtains (2.4). If $|S| \geq 4$, the new total is negative. When $|S| = 3$, all block caps are inactive and $C_S(\ell) = \binom{\ell + n - 4}{n - 1}$, whose numerator is z^3 . This proves (2.5)–(2.6). $\square$

3 The binomial-tail kernel

For a decreasing sequence $p = (p_1, p_2, \dots)$ with

$$p_1 \leq \alpha, \quad p_i \geq 0, \quad \sum_i p_i \leq 1,$$

write $d = 1 - \sum_i p_i$ for its dust mass and define

$$B_m(p) = 1 - \sum_i \sum_{j=m+1}^{2m} \binom{2m}{j} p_i^j (1 - p_i)^{2m-j}, \quad B_0 = 1. \quad (3.1)$$

For a finite probability vector this is the probability that no class receives more than m balls in a multinomial experiment with $2m$ trials. Dust is treated as an uncapped class. Set

$$G_p(t) = \sum_{m \geq 0} B_m(p) \frac{t^m}{(2m)!}, \quad F_p(z) = \sum_{k \geq 0} B_{2k}(p) \frac{z^k}{(4k)!}. \quad (3.2)$$

Let

$$\mathcal{P}_\alpha^* = \{p_1 \geq p_2 \geq \cdots \geq 0 : p_1 \leq \alpha, \sum_i p_i \leq 1\}. \quad (3.3)$$

This space is compact in the product topology. The estimate

$$\sum_{i > R} \Pr\{\text{Bin}(2m, p_i) > m\} \leq 2^{2m} (R+1)^{-m} \quad (3.4)$$

shows that B_m , hence G_p and F_p on compact subsets of the complex plane, varies continuously with p .

The analytic input is the following theorem.

Theorem 3.1 (positive-real kernel). *For every $p \in \mathcal{P}_\alpha^*$,*

$$\Re \frac{G_p(t)}{\cosh \sqrt{t}} > 0 \quad (\Re t \geq 0). \quad (3.5)$$

Consequently G_p is Hurwitz stable, and F_p has infinitely many simple, strictly negative real zeros.

Proof. We give the coefficient gate in detail because its uniformity is essential. Put $N = 4k$, $T = 2^{N/2}$, and, for one capped class, define

$$C_k(x) = (-1)^k \sum_{m=1}^{2k} (-1)^m \binom{N}{2m} \sum_{j=m+1}^{2m} \binom{2m}{j} x^j (1-x)^{2m-j}. \quad (3.6)$$

Write its degree- N Bernstein expansion as

$$C_k((1-h)/2) = \sum_{\ell=0}^N U_k(\ell) \binom{N}{\ell} h^\ell (1-h)^{N-\ell}. \quad (3.7)$$

The scalar gate needed below is

$$U_k(\ell) \leq \left(1 - \frac{\ell}{N}\right) U_k(0), \quad 0 \leq \ell \leq N. \quad (3.8)$$

We now prove (3.8). Let

$$S(u) = (-1)^k (2-u)^{N/2} P_N((2-u)^{-1/2}), \quad B = S(0), \quad D_0 = S'(0), \quad (3.9)$$

where P_N is the Legendre polynomial. Direct differentiation gives

$$\frac{d}{dh} C_k((1-h)/2) = (1-h) S'(h^2), \quad U_k(0) = \frac{T-B}{2}. \quad (3.10)$$

The Legendre equation, written in Prüfer coordinates between $\theta = \pi/4$ and $\pi/2$, gives the non-asymptotic energy estimate

$$-D_0 \geq \frac{T\sqrt{N}}{4}. \quad (3.11)$$

If $S'(h^2) = \sum_{j \geq 0} a_j h^{2j}$, the differentiated hypergeometric equation gives

$$2(j+2)(j+1)a_{j+2} + (j+1)(N-11/2-3j)a_{j+1} + (j-(N-2)/2)(j-(N-3)/2)a_j = 0. \quad (3.12)$$

Besides $a_0 = D_0$, direct coefficient comparison gives

$$\frac{a_1}{D_0} = -\frac{N-5/2}{2} + \frac{N(N-1)B}{8(-D_0)}.$$

The same energy estimate implies $-D_0 \geq NB/3$ for $N \geq 8$, and hence $-N/2 < a_1/D_0 \leq (7-N)/8 \leq 0$. Thus $|a_1| < N(-D_0)/2$; the remaining case $N = 4$ has $a_1 = -3/4$ and satisfies the same absolute bound. Starting from these two initial coefficients, induction in (3.12) yields

$$|a_j| \leq (-D_0) \frac{(3N/4)^j}{j!}. \quad (3.13)$$

After conversion back to Bernstein increments, (3.13) proves (3.8) for

$$0 \leq \ell \leq \left\lfloor \frac{2(N-2)}{3\sqrt{N}} \right\rfloor. \quad (3.14)$$

For the remaining prefix put $e = N^{-1/2}$, $a = \ell N^{-1/2}$, $v = e^2$, $\rho = NB/(-D_0)$, and $b_j = a_j j! / (D_0 N^j)$. Equation (3.12) becomes

$$\begin{aligned} b_0 &= 1, \\ b_1 &= -\frac{1}{2} + \frac{5}{4}v + \frac{1}{8}(1-v)\rho, \\ b_{j+2} &= -\frac{1}{2}\{[1 - (3j+11/2)v]b_{j+1} + [1/4 - (j+5/4)v + (j+1)(j+3/2)v^2]b_j\}. \end{aligned} \quad (3.15)$$

The same Prüfer phase gives

$$\frac{4}{1 + (\sqrt{2}-1) + (\sqrt{2}-1)/(2N)} \leq \rho \leq \frac{4}{1 + (\sqrt{2}-1) + (\sqrt{2}-2)/(2N)}. \quad (3.16)$$

Integrating (3.10) and converting to Bernstein form gives

$$U_k(0) - U_k(\ell) = (-D_0) \sum_j b_j \frac{N^j}{j!} Q_j, \quad (3.17)$$

where

$$Q_j = \frac{\binom{\ell}{2j+1}}{(2j+1)\binom{N}{2j+1}} - \frac{\binom{\ell}{2j+2}}{(2j+2)\binom{N}{2j+2}}. \quad (3.18)$$

For $N \geq 256$ and $13\sqrt{N}/20 \leq \ell \leq 49\sqrt{N}/16$, outward rational interval evaluation of (3.15)–(3.18), followed by

$$\text{Tail}_J \leq \frac{x^{J+1}}{(J+1)!(2J+3)} \frac{1}{1-x/(J+2)}, \quad x = \frac{3a^2}{4(1-2v)^2}, \quad (3.19)$$

proves $U_k(0) - U_k(\ell) > (-D_0)\ell/(2N)$. The 2403 certified interval leaves cover this entire compact rectangle, not a point grid. Together with (3.11) and the overlap in (3.14), this closes the prefix through $\ell_0 = \lceil 3\sqrt{N} \rceil$.

For $\ell_0 \leq \ell \leq N/3$, let $\text{Pos}(f)$ denote the sum of the strictly positive-power coefficients of a finite Laurent polynomial f . The exact Laurent representation

$$U_k(\ell) = (-1)^k \Re \text{Pos} \left[\left(1 + \frac{i}{2}(z + z^{-1}) \right)^{N-\ell} (1 + i/z)^\ell \right] \quad (3.20)$$

and a coefficient Cauchy integral give

$$|U_k(s+1) - U_k(s)| \leq \frac{5T}{8\sqrt{N}} e^{-3s^2/(16N)}. \quad (3.21)$$

The tail beyond $3\sqrt{N}$ is less than $T/8$, so $U_k(0) - U_k(\ell) > T/4 > (\ell/N)U_k(0)$ on this band. For $\ell \geq \lceil N/3 \rceil$, write $r = N - \ell$, $K(z) = 1 + i(z + z^{-1})/2$, and $L(z) = 1 + i/z$. Cauchy's formula gives

$$|U_k(\ell)| \leq \frac{\max_{|z|=R} |K(z)|^r |L(z)|^\ell}{R-1}. \quad (3.20a)$$

On each of sixteen rational subintervals of $s = \sin(\arg z) \in [-1, 1]$, exact Bernstein certificates prove

$$A^2B < 7, \quad AB^3 < 13 \quad (R = 8/5), \quad AB^3 < 8 \quad (R = 3), \quad (3.20b)$$

where $A = |K|^2$ and $B = |L|^2$. Interpolation in $\lambda = \ell/N$ therefore yields, for $N \geq 160$,

$$|U_k(\ell)| \leq \frac{5}{3}T \max\{(7/8)^{N/6}, (13/16)^{N/8}\} < \frac{T}{16} \quad (N/3 \leq \ell \leq 3N/4),$$

and

$$|U_k(\ell)| \leq \frac{1}{2}T(8/9)^{N/2} < \frac{T}{2N} \quad (3N/4 \leq \ell \leq N-2).$$

Together with $T/4 < U_k(0) < T/2$, these are (3.8); the last two indices $\ell = N-1, N$ vanish exactly. For $N = 4, 8, \dots, 156$, the remaining 2028 pairs $\lceil N/3 \rceil \leq \ell \leq N-2$ are closed by an independent integer Laurent-recurrence certificate, including the three continuous certificates in (3.20b). Separately, the remaining $N < 256$, $1 \leq \ell < \lceil N/3 \rceil$ comprise exactly 2646 integer pairs; a second certificate reconstructs their Bernstein numerators from (3.12) and checks (3.8) exactly. The analytic bands and these two finite complements cover every index and overlap at their joins.

Writing $h = 1 - 2x$, (3.8) gives

$$C_k(x) \leq U_k(0) \sum_{\ell=0}^N \left(1 - \frac{\ell}{N}\right) \binom{N}{\ell} h^\ell (1-h)^{N-\ell} = 2xU_k(0). \quad (3.22)$$

Two classes cannot both receive more than half of the trials. Therefore the bad events are pairwise disjoint, and summing (3.22) over all classes uses only $\sum_i p_i \leq 1$. The exact product identity on the boundary ray $w = x(1+i)$ then yields

$$\Re(G_p(w^2)\overline{\cosh w}) = 1 + \sum_{k \geq 1} A_k(p) \frac{4^k x^{4k}}{(4k)!} > 0. \quad (3.23)$$

Indeed,

$$A_k(p) \geq B_k := (-1)^k 2^{2k} P_{4k}(1/\sqrt{2}) > 0. \quad (3.24)$$

For the last sign, the continuous Legendre Prüfer phase at $\theta = \pi/4$ lies strictly between $k\pi + \pi/8$ and $k\pi + 3\pi/8$.

To pass from the boundary to the half-plane, use the signed arcsine measures μ_{p_i} for the binomial tails. They give the exact identity

$$G_p(w^2) = \cosh w - \sum_i \int \cosh(w\sqrt{x}) d\mu_{p_i}(x), \quad \text{supp } \mu_{p_i} \subseteq [0, 4p_i(1-p_i)]. \quad (3.25a)$$

Their total variations are

$$v(x) = 1 - \frac{4}{\pi} \arctan \sqrt{1-2x}, \quad 0 \leq x \leq 1/2. \quad (3.25)$$

The function v is convex, $v(0) = 0$, $v(1/2) = 1$, and $v(x) \leq 2x$. Thus $\sum_i v(p_i) \leq 2$, including countably many coordinates. If $|\arg w| \leq \pi/4$, then $|\cosh(aw)| \leq |\cosh w|$ for $0 \leq a \leq 1$; the support bound in (3.25a) has $\sqrt{x} \leq 1$. Consequently the quotient in (3.5) is analytic and bounded in the right half-plane, by three in modulus. Its boundary real part is strictly positive by (3.23). Mapping the half-plane to the disk and applying the Poisson formula, with the point at infinity having zero harmonic measure, proves (3.5).

Since $G_p(0) = 1$, has order at most $1/2$, and is not a polynomial, its genus-zero product has all zeros in the open left half-plane. For every finite canonical-product truncation, Hermite–Biehler places the zeros of its even part on the negative real axis and interlaces them with those of the odd part. The products and their first derivatives converge uniformly on compact sets, so Hurwitz passes the location statement to the limit. If $\theta_p(y) = \arg G_p(iy)$, the product gives

$$\theta'_p(y) = \sum_j \Re \frac{1}{\zeta_j + iy} > 0. \quad (3.25b)$$

where $-\zeta_j$ are the zeros of G_p . Because $F_p(-y^2) = \Re G_p(iy)$, every zero is simple. Positivity of the constant term excludes zero. $\square$

Moreover, if $r = \sqrt{4\alpha(1-\alpha)} < 1$ and $u = \Re w > 0$, then

$$\left| \frac{G_p(w^2)}{\cosh w} - 1 \right| \leq \frac{4e^{-(1-r)u}}{1 - e^{-2u}}. \quad (3.26)$$

It follows that the phase and any fixed initial collection of zeros of F_p are uniform over $\mathcal{P}_\alpha^*$. More precisely, take the continuous phase

$$\theta_p(y) = \arg G_p(iy), \quad \theta_p(0) = 0.$$

The bound (3.26) gives, uniformly in p ,

$$\theta_p(T^2) = T/\sqrt{2} + o(1). \quad (3.27)$$

Consequently, for every sufficiently large fixed integer j , the radius $T_j = \sqrt{2\pi}j$ contains exactly j zeros of F_p in $(-T_j^4, 0)$, and

$$\text{sign } F_p(-T_j^4) = (-1)^j. \quad (3.28)$$

This is an exact root count, not merely the existence of j negative roots.

4 The two endpoint limits

Let $c_0 = \lceil n/4 \rceil$, $b = 4c_0 - n \in \{0, 1, 2, 3\}$, and $D = n - c_0$.

Lemma 4.1 (small-root limit). *If $p^{(n)} = (m_1/n, m_2/n, \dots)$, padded by zeros and ordered decreasingly, then, uniformly over all allowed partitions,*

$$H_{\mathbf{m}}(z/n^4) - F_{p^{(n)}}(z) \longrightarrow 0 \quad (4.1)$$

on every compact subset of $\mathbb{C}$.

Proof. For fixed k , the leading $0/1$ contribution counts allocations of $4k$ distinct elements with no block used more than $2k$ times. Sampling with replacement differs from sampling distinct elements with probability $O_k(n^{-1})$, uniformly in the partition. Contributions supported on at most $4k - 1$ elements are $O_k(n^{4k-1})$. The lower Ehrhart-to- h^* convolution terms disappear after normalization, while nonnegativity and removal of all constraints gives a common factorial majorant. Coefficientwise convergence therefore upgrades to (4.1). $\square$

Fix a sufficiently large J . For each parameter p , isolate the first J simple roots from (3.28) by disjoint conjugation-invariant circles and choose an outer circle on which F_p is nonzero. Compact convergence and simple-root continuity preserve these circles on a neighborhood of p . Finitely many such neighborhoods cover the compact space $\mathcal{P}_\alpha^*$. Applying (4.1) and Rouché's theorem on each member of this finite cover shows that, for all sufficiently large n , $H_{\mathbf{m}}$ has exactly J roots in the corresponding scaled initial region. Conjugation symmetry and the parameter-local isolated real roots force the actual roots to be simple and negative. A single outer circle, obtained from the uniform far-right bound (3.26), gives the common root count and the fixed initial grid point inherits the sign $(-1)^J$.

Lemma 4.2 (reciprocal endpoint). *Define*

$$Q_n(z) = n^{-b}(z/n^4)^D H_{\mathbf{m}}(n^4/z).$$

Uniformly over all allowed partitions,

$$Q_n(z) \longrightarrow V_{4,b}(z) := \sum_{k \geq 0} \frac{z^k}{(4k+b)!} \quad (4.2)$$

on compact subsets of $\mathbb{C}$. Each $V_{4,b}$ has infinitely many simple, strictly negative zeros. This is the residue function E_b of Adiprasito–Zhang [7, Section 3.3], with $k = 4$; its zero property is prior work. The uniform convergence over capacity-two partitions is the assertion needed here. The multiplier-sequence argument below records an alternative derivation of the known zero property, not a novelty claim.

Proof. At dilation $c_0 + k$, subtract one from every interior coordinate. For fixed k , the residual total is $b + 4k$, whereas every coordinate cap and every block cap tends uniformly to infinity because $\max_i m_i \leq \alpha n$ with $\alpha < 1/2$. Thus the interior lattice-point count is exactly $\binom{n+b+4k-1}{b+4k}$ for all sufficiently large n , uniformly in the partition. Ehrhart reciprocity and a factorial majorant give (4.2).

The gamma multiplication formula writes

$$V_{4,b}(z) = \frac{1}{b!} \sum_{k \geq 0} \frac{(z/256)^k}{k! \prod_{\gamma \in A_b} (\gamma)_k}, \quad A_b = \left\{ \frac{b+1}{4}, \frac{b+2}{4}, \frac{b+3}{4}, \frac{b+4}{4} \right\} \setminus \{1\}. \quad (4.3)$$

Every remaining parameter is positive, and the reciprocal Pochhammer sequences are classical nonnegative multiplier sequences. Hence $V_{4,b}$ belongs to the Laguerre–Pólya class. To exclude multiplicities, retain one factor $1/(\gamma)_k$ until last and shift it to $1/(\gamma + m)_k$. Reversing that shift applies successively the positive Euler operators $1 + \vartheta/(\gamma + r)$, each of which reduces the multiplicity of every nonzero old root by one. The shifted inputs converge locally to a positive constant as $m \rightarrow \infty$, so a fixed multiple root is impossible. The positive constant term excludes the origin.

The signs and counts used below are exact. Put

$$T_{k,b} = \sqrt{2}\pi(k + b/4).$$

Apply the four-root-of-unity filter on the sector $0 \leq \arg w \leq \pi/4$ and write

$$W(w) := 4e^{-w}w^b V_{4,b}(w^4) = 1 + \lambda(w) + E(w),$$

where

$$\lambda(w) = e^{2ib\pi/4} e^{w(e^{-i\pi/2}-1)}$$

and the two remaining root-of-unity terms form $E(w) = O(e^{-cT_{k,b}})$, uniformly on the half-sector. Put $w = T_{k,b}e^{i\psi}$ and $\delta = \pi/4 - \psi$. Then

$$|\lambda| = e^{-\sqrt{2}T_{k,b} \sin \delta}, \quad \arg \lambda = -2k\pi + \sqrt{2}T_{k,b}(1 - \cos \delta).$$

When $\delta \leq T_{k,b}^{-2/3}$, the angle between 1 and λ tends uniformly to zero; on the complementary interval $|\lambda|$ is exponentially small. Hence, for all sufficiently large k , $\Re W > 1/2$ on the entire half-sector. In particular the boundary has no zero and the continuous argument of W returns to zero at the two endpoints. The factor $4e^{-w}w^b$ contributes exactly $-k\pi$ on the upper semicircle, so $V_{4,b}(w^4)$ contributes $k\pi$; conjugation gives the same increment below. The argument principle therefore gives

$$\#\{z : V_{4,b}(z) = 0, |z| < T_{k,b}^4\} = k, \quad \text{sign } V_{4,b}(-T_{k,b}^4) = (-1)^k. \quad (4.4)$$

At $w = T_{k,b}e^{i\pi/4}$, both endpoint factors are real, and the same continuous-argument computation gives the sign in (4.4). The four values of b have a common lower bound on k . $\square$

For each fixed K beyond this common bound, the relation

$$\phi(t) = \frac{3\pi}{4} - \frac{1}{\sqrt{2}t} + O(t^{-2})$$

implies $n/t_{n,D-K} \rightarrow T_{K,b}$. Compact convergence in (4.2) on the corresponding moving outer circle and (4.4) therefore transfer exactly K roots to the reciprocal end of $H_{\mathbf{m}}$, again simple and negative. At the terminal grid point,

$$\text{sign } H_{\mathbf{m}}(-t_{n,D-K}^4) = (-1)^{D-K}. \quad (4.5)$$

The two lemmas give fixed, uniform numbers of simple negative roots near the two ends of the negative axis. It remains to fill the moving middle without losing uniformity when q grows.

5 A block-count-free moving-cap estimate

Put $x = -t^4$, $A = 1 + t^4$, and let $W_{n-1}^{(2,2)}(u, x)$ denote the two-marker transfer polynomial arising from one active block cap in (2.6). For a block of size h , its cumulative cap term is

$$K_h = \frac{1}{2\pi i} \int_{|u|=\rho} W_{n-1}^{(2,2)}(u, -t^4) \frac{u^{-h}}{1-u} du. \quad (5.1)$$

Lemma 5.1 (uniform moving caps). *For every fixed $\alpha < 1/2$, every partition with $\sum_i h_i = n$ and $1 \leq h_i \leq \alpha n$, and every sequence $t \rightarrow 0$ with $T = nt \rightarrow \infty$,*

$$\frac{\sum_i |K_{h_i}|}{R_4(t)^n} \rightarrow 0, \quad (5.2)$$

where $R_4(t)$ is the dominant four-color spectral radius.

Proof. The integrals of $W/(1-u)$ and of the zero-jump term $A^n u^{n-1}/(1-u)$ vanish. Therefore

$$K_h = \frac{1}{2\pi i} \int (W - A^n u^{n-1}) \frac{u^{-h} - 1}{1-u} du. \quad (5.3)$$

With $\rho = e^{-\eta t}$,

$$\sum_i \left| \frac{u^{-h_i} - 1}{1-u} \right| \leq n\rho^{-\alpha n}. \quad (5.4)$$

On the near arc $|\arg u| \leq Lt$, the Schur expansion gives

$$\frac{|W - A^n u^{n-1}|}{R_4^n} \leq C_L(1+T)^3 e^{-\beta \eta T} + C e^{-c_\star T}, \quad (5.5)$$

for $\alpha < \beta < 1/2$, where one may take the fixed spectral-gap constant $c_\star = 1/(4\sqrt{2})$. Equations (5.4)–(5.5) are exponentially small after integration.

On the complementary arc, the two spectral branches solve

$$(\lambda - A)(\lambda - Au) = z\lambda^2, \quad z = it^2 \text{ or } -it^2. \quad (5.6)$$

They are analytic about $z = 0$. Taylor expansion of the paired values at $\pm it^2$ cancels the linear term. Uniform implicit-derivative bounds give

$$\frac{|W - A^n u^{n-1}|}{R_4^n} \leq Ct^4(\delta^{-4} + n\delta^{-3} + n^2\delta^{-2})e^{-c_\star T}, \quad \delta = |1-u|. \quad (5.7)$$

After (5.4) and integration, the far contribution is at most

$$C_L(T + T^2 + T^3)e^{-(c_\star - \eta\alpha)T}.$$

Choose $\eta\alpha < c_\star$. This proves (5.2). The cancellation between the two signs in (5.6), together with $\sum_i h_i = n$, is what removes any factor depending on the number of blocks. $\square$

6 Uniform signs on the complete root grid

Let

$$\mu(t) = \frac{1 + t^4}{1 - te^{\pi i/4}} = R_4(t)e^{i\phi(t)}.$$

The phase ϕ increases strictly from 0 to $3\pi/4$. Define the grid $t_{n,j} > 0$ by

$$n\phi(t_{n,j}) = j\pi. \tag{6.1}$$

The two-branch phase grid and its assembly with direct and reciprocal endpoints follow the ordinary-hypersimplex architecture of [7, Sections 3.4–3.5]. The partition corrections, cancellation uniform in the number of blocks, and compactification with dust provide the additional controls for the present family.

Proposition 6.1 (middle-grid signs). *For every fixed $\alpha < 1/2$, there are fixed integers J, K and a common threshold such that, for all allowed partitions and $J \leq j \leq D - K$, the signed values of $H_{\mathbf{m}}(-t_{n,j}^4)/R_4(t_{n,j})^n$ alternate with j and are nonzero.*

Proof. Use (2.6) and split any hypothetical failure sequence according to the limit of $t_{n,j}$.

On a fixed compact subinterval of $(0, \infty)$, the four-color term has a uniform spectral gap. Each highest cap is exponentially small; summing at most n of them is harmless. The lower color levels carry the multiplicities $e_s(\mathbf{m}) \leq n^s/s!$, $s = 1, 2, 3$, and within each C_S there are at most $q \leq n$ single-cap corrections. Thus their total prefactor is at worst $O(n^4)$, which is still absorbed by the uniform spectral gap.

If $t \rightarrow \infty$ while $n/t \rightarrow \infty$, the same transfer spectrum gives the large-end moving estimate; polynomial factors from the first three color levels are absorbed by the exponential gap. If $t \rightarrow 0$ while $nt \rightarrow \infty$, the highest caps are controlled by Lemma 5.1, the lower levels by CT^se^{-cT} , and the three-color term by $CT^3t^9e^{-cT}$. In all three moving regimes, the normalized signed value tends to $1/2$.

The remaining possibilities have bounded initial index or bounded terminal index. They are excluded respectively by Lemma 4.1 together with Theorem 3.1, and by Lemma 4.2. Compactness of $\mathcal{P}_\alpha^*$ allows passage to a convergent subsequence even when the number of blocks diverges or mass turns into dust. The fixed endpoint values need only retain the nonzero signs supplied by their kernels; they are not asserted to converge to $1/2$. This exhausts every failure sequence. $\square$

7 Proof of Theorem 1.1

Choose J, K large enough that the two endpoint kernels give, uniformly over $\mathcal{P}_\alpha^*$, exactly J simple negative roots at the small-root end and exactly K simple negative roots at the reciprocal end. Increase the common threshold so that Proposition 6.1 applies.

The consecutive grid signs in the middle yield $D - J - K$ distinct negative roots. Together with the endpoint roots this gives

$$J + (D - J - K) + K = D$$

distinct negative roots. Lemma 2.1 says that $H_{\mathbf{m}}$ has degree exactly D ; hence these roots exhaust the full complex zero set and all are simple. $\square$

8 Reproducibility

The frozen prewrite proof package consists of 37 source and certificate files. Its path-and-SHA-256 manifest is

`loops/NONPAVING-HSTAR-PUBLICATION-PREP-0001/A1-证明依赖闭包.sha256`.

The finite scalar certificates supporting the gate (3.8) are checked in exact rational arithmetic. One closes the 2028 large-index pairs $N = 4, 8, \dots, 156$, together with the three continuous Bernstein certificates in (3.20b); the other closes 2403 interval boxes and the 2646 small-order prefix pairs. The public package, if released, must use the non-optimized project interpreter because the checkers deliberately use assertions, and must include hard input-size gates and the frozen certificate digests. These finite calculations close stated finite complements inside a general proof; they are not empirical evidence for Theorem 1.1.

9 Prior work and limitations

Lam–Postnikov weighted multi-hypersimplices, elementary split matroids, cuspidal matroids, and subsequent multi-hypersimplex formulas supply object descriptions and Ehrhart-counting precedents. We do not claim these as new. The sufficient conditions stated in Li’s arXiv v1 for zonotopal and saturated settings do not, as currently stated, automatically cover all polytopes in (1.1), and they do not give the simple-root conclusion used here.

Adiprasito and Zhang [7, Theorem 3.1] prove eventual simple negative real-rootedness for ordinary hypersimplices of each fixed rank $k \geq 3$. Its $k = 4$ specialization includes the uniform degeneration of our family. We checked the full text of HAL version v1: it contains no weighted multi-hypersimplex or general perturbation theorem covering the nonuniform partitions of Theorem 1.1. The increment relative to that rank-four specialization is the common threshold over all capacity-two partitions of bounded maximum block ratio. This does not assert a theorem for other ranks, or establish global priority over all other or unindexed work. The residue-function and phase-grid prototypes in its Sections 3.3–3.5 are explicitly treated as prior mechanisms, not as new results here.

The following are not proved here:

- the critical boundary $\max_i m_i = n/2$;
- a common theorem for every small n ;
- an explicit or α -uniform threshold;
- other ranks, block capacities, or arbitrary nonpaving matroids;
- any global priority claim.

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AI-assisted research disclosure

The research, proof development, checking, and writing were conducted with substantial AI assistance. Carptopus is the human author and assumes responsibility for the mathematical claims and any released version.